

ENG 1005 Quick Recap

Scalar : Quantities represented by a numerical value

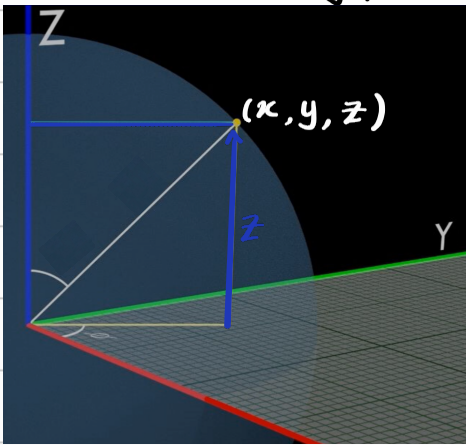
Scalar Field : Func. that assigns a scalar to every point in a region

Vector : Scalar with direction

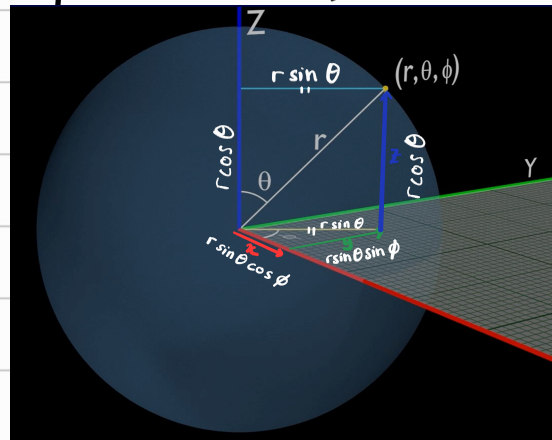
Vector Field : Func. that assigns a vector to every point in a region

Coordinate Systems

① Cartesian (x, y, z)

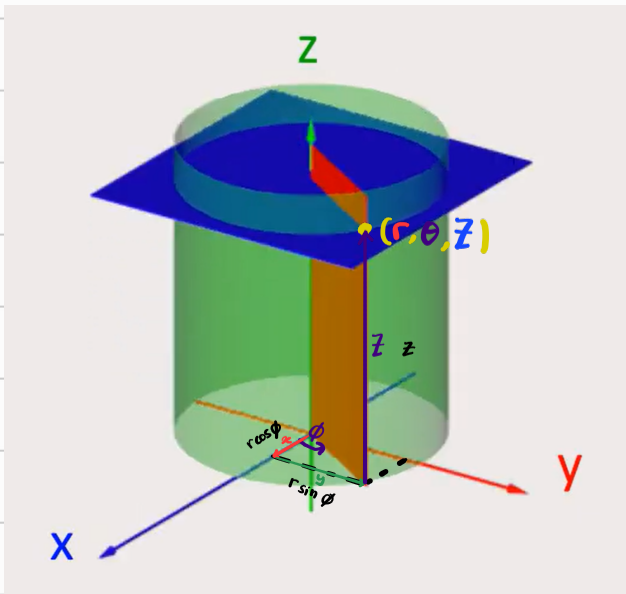


② Spherical (r, θ, ϕ)



$$\begin{aligned} x &= r \sin \theta \cos \phi \\ y &= r \sin \theta \sin \phi \\ z &= r \cos \theta \end{aligned}$$

③ Cylindrical (ρ, ϕ, z)



$$\begin{aligned} x &= \rho \cos \phi \\ y &= \rho \sin \phi \\ z &= z \end{aligned}$$

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Dot Product : Multiplication of 2 vectors magnitude

[Projecting one vector onto another]

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix} = 2a + 3b + 4c \quad (\text{OR}) \quad a \cdot b = |a||b|\cos\theta$$

Cross Product : To find perpendicular vector to 2 vectors

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} \times \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 4b - 3c \\ -(4a - 2c) \\ 3a - 2b \end{pmatrix} \quad (\text{OR}) \quad a \times b = |a||b|\hat{n}\sin\theta$$

Vectors

[More In - Depth]

In Cartesian, Let $\vec{V} = \begin{bmatrix} 3 \\ 4 \end{bmatrix} = 3\hat{i} + 4\hat{j}$ Components

In ECE 2111, the notation is $\vec{V} = 3\hat{a}_x + 4\hat{a}_y$ Basis Vectors

This means $\vec{V} = 3[\text{x-dirac unit vec.}] + 4[\text{y-dirac unit vec.}]$

Generally, every vector are just sums of scaled basis vectors.

Alternately, every vector are a sum of the contributions of each basis vector

Components

$\text{Vector} = A\hat{a}_1 + B\hat{a}_2 + C\hat{a}_3 + \dots$

Basis Vectors

Coordinate System	Coordinate	Basis Vectors, $\hat{a}$
Cartesian	(x, y, z)	$\hat{a}_x, \hat{a}_y, \hat{a}_z$
Cylindrical	(ρ, ϕ, z)	$\hat{a}_\rho, \hat{a}_\phi, \hat{a}_z$
Spherical	(r, θ, ϕ)	$\hat{a}_r, \hat{a}_\theta, \hat{a}_\phi$

Misconception Clarification

Coordinates : SCALAR values to identify a point

Basis Vector : Unit VECTOR giving the current local coordinate direc.

The coordinate system defines the grid ; the basis vec. come from that grid

↳ Basis vector are straight unit tangent arrows to the coordinate grid lines at a particular point

Mental Model

We should NOT visualise $(4, 45^\circ)$ as

"Go 4 units along $+x$, move around circle until $\phi = 45^\circ$ "

$(4, 45^\circ)$ is the coordinate, its "your final location" not "the path you used to get there"

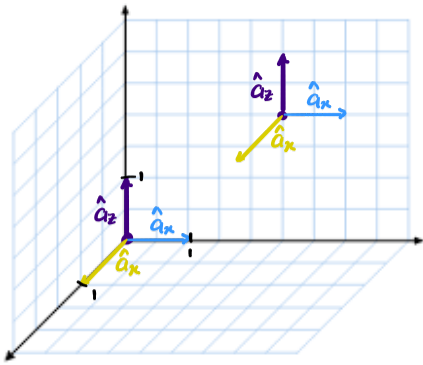
As a matter of fact, there are infinite # of paths you can take to ultimately reach the coordinate $(4, 45^\circ)$

Thus, coordinates only specify the final location. To describe the path to that final location, we introduce the basis vectors of that coordinate system.

Basis Vectors, $\hat{a}$

Basis vector are straight unit tangent arrows to the coordinate grid lines at a particular point

Cartesian



Note that $\hat{a}_x, \hat{a}_y, \hat{a}_z$ point in the same direction at every point of the grid

Change of Basis Formula

Cylindrical $\rightarrow$ Cartesian

$$\hat{a}_x = \cos \phi \hat{a}_\rho - \sin \phi \hat{a}_\phi$$

$$\hat{a}_y = \sin \phi \hat{a}_\rho + \cos \phi \hat{a}_\phi$$

$$\hat{a}_z = \hat{a}_z$$

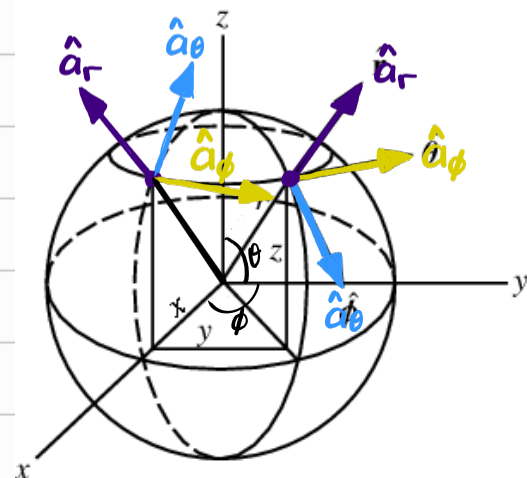
Spherical $\rightarrow$ Cartesian

$$\hat{a}_x = \cos \phi [\sin \theta \hat{a}_r + \cos \theta \hat{a}_\theta] - \sin \phi \hat{a}_\phi$$

$$\hat{a}_y = \sin \phi [\sin \theta \hat{a}_r + \cos \theta \hat{a}_\theta] + \cos \phi \hat{a}_\phi$$

$$\hat{a}_z = \cos \theta \hat{a}_r - \sin \theta \hat{a}_\theta$$

Spherical



Note that $\hat{a}_r, \hat{a}_\theta, \hat{a}_\phi$ directions vary from one point to another

Change of Basis Formula

Cartesian $\rightarrow$ Spherical

$$\hat{a}_r = \sin \theta [\cos \phi \hat{a}_x + \sin \phi \hat{a}_y] + \cos \theta \hat{a}_z$$

$$\hat{a}_\theta = \cos \theta [\cos \phi \hat{a}_x + \sin \phi \hat{a}_y] - \sin \theta \hat{a}_z$$

$$\hat{a}_\phi = -\sin \phi \hat{a}_x + \cos \phi \hat{a}_y$$

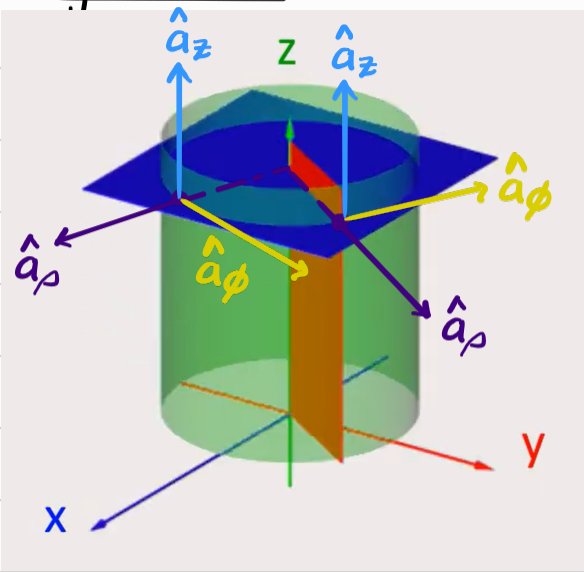
Cylindrical $\rightarrow$ Spherical

$$\hat{a}_r = \sin \theta \hat{a}_\rho + \cos \theta \hat{a}_z$$

$$\hat{a}_\theta = \cos \theta \hat{a}_\rho - \sin \theta \hat{a}_z$$

$$\hat{a}_\phi = \hat{a}_\phi$$

Cylindrical



Note that $\hat{a}_\rho, \hat{a}_\phi, \hat{a}_z$ directions vary from one point to another

Change of Basis Formula

Cartesian $\rightarrow$ Cylindrical

$$\hat{a}_\rho = \cos \phi \hat{a}_x + \sin \phi \hat{a}_y$$

$$\hat{a}_\phi = -\sin \phi \hat{a}_x + \cos \phi \hat{a}_y$$

$$\hat{a}_z = \hat{a}_z$$

Spherical $\rightarrow$ Cylindrical

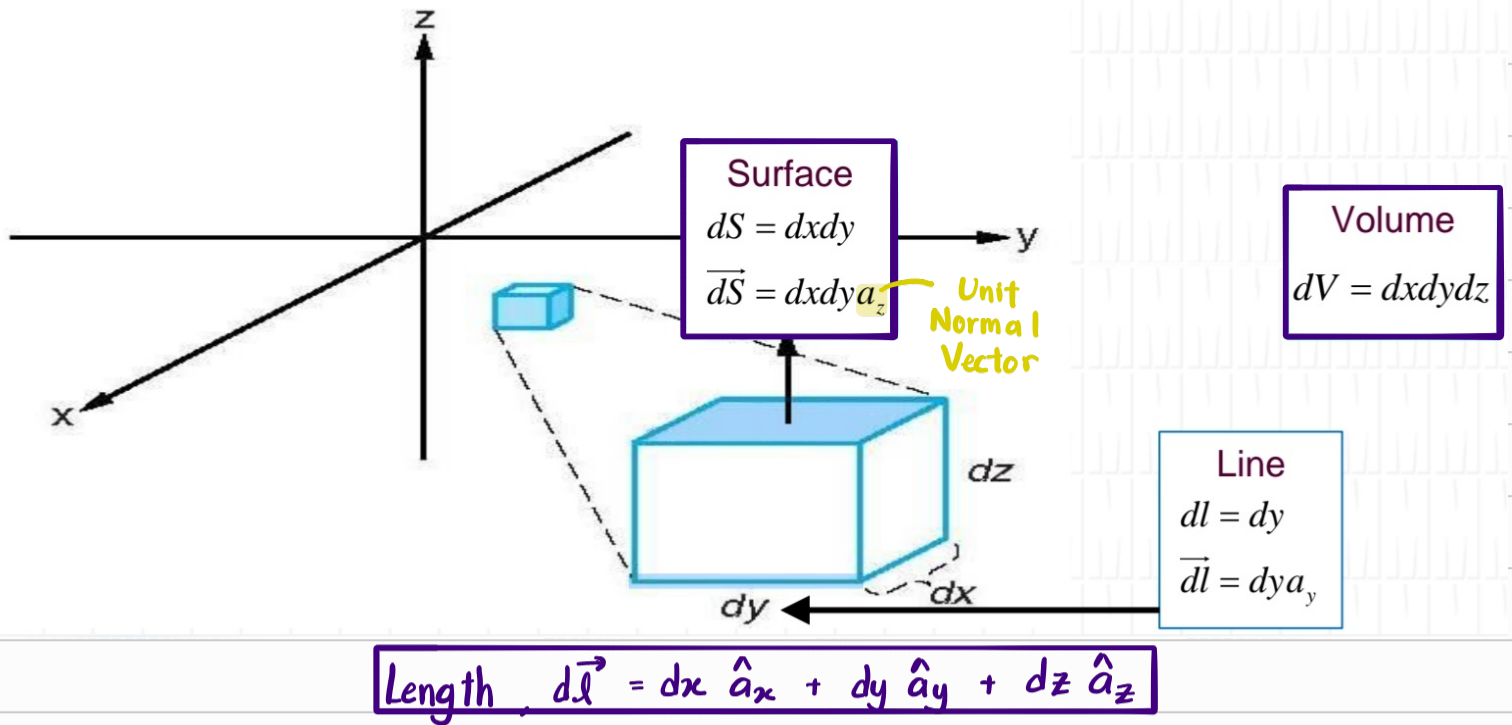
$$\hat{a}_\rho = \sin \theta \hat{a}_r + \cos \theta \hat{a}_\theta$$

$$\hat{a}_\phi = \hat{a}_\phi$$

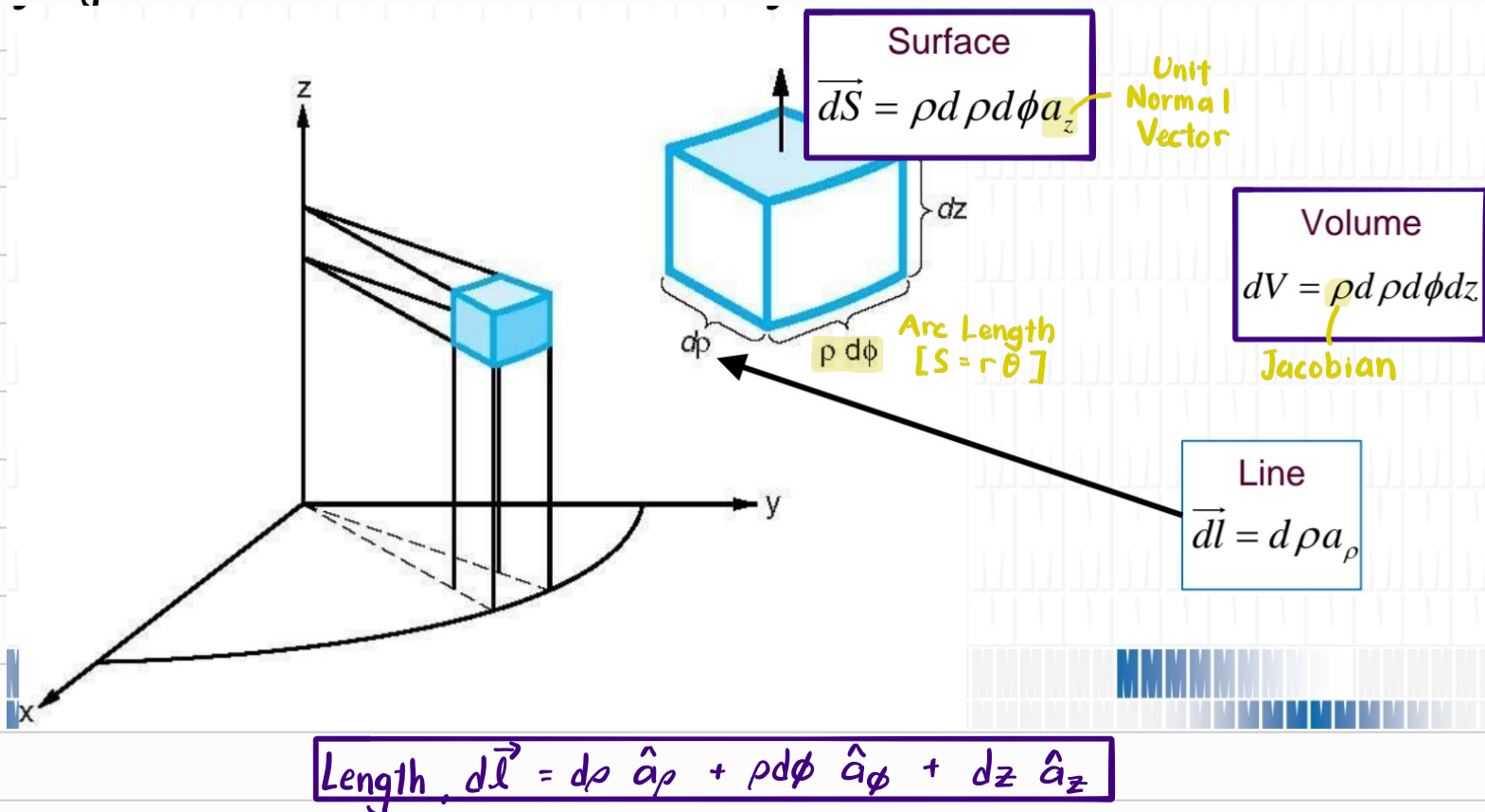
$$\hat{a}_z = \cos \theta \hat{a}_r - \sin \theta \hat{a}_\theta$$

Differential Length, Area, Volume

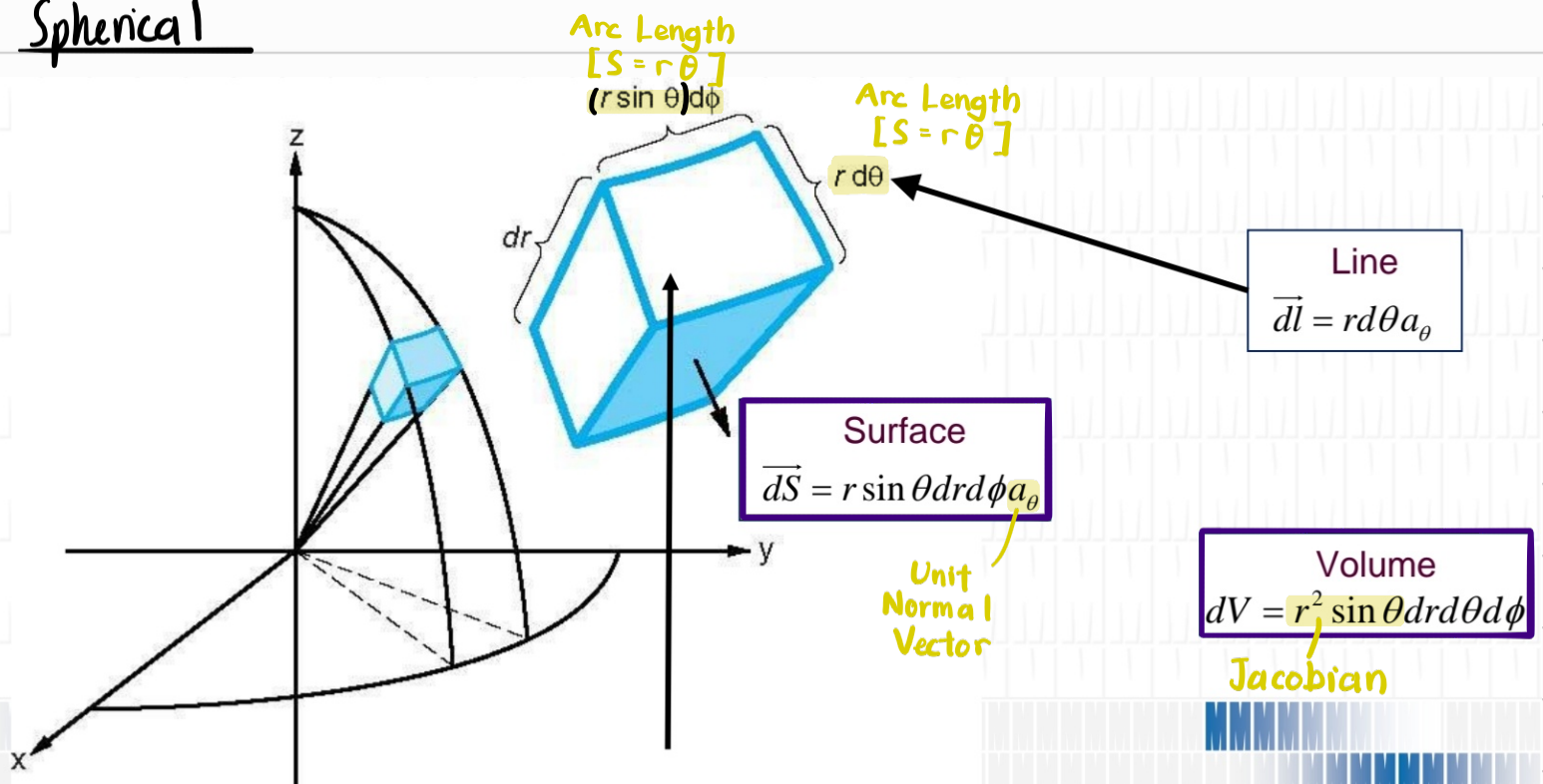
Cartesian



Cylindrical

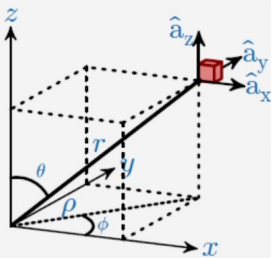


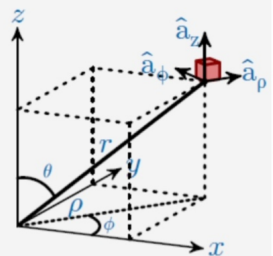
Spherical

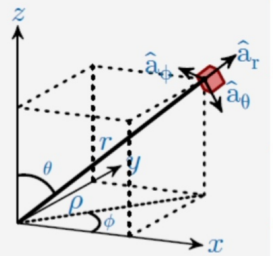


$$\text{Length, } d\vec{l} = dr \hat{a}_r + r d\theta \hat{a}_\theta + (r \sin \theta) d\phi \hat{a}_\phi$$

Quick Glance Tables

Cartesian		x	y	z
 $\vec{r} = (x, y, z)$		$\rho \cos \phi$	$\rho \sin \phi$	z
	$\oplus$	$r \sin \theta \cos \phi$	$r \sin \theta \sin \phi$	$r \cos \theta$
	$d\vec{\ell}$	$dx\hat{a}_x + dy\hat{a}_y + dz\hat{a}_z$		
	dV	$dx dy dz$		
		Cylindrical	Spherical	
	$\hat{a}_x$	$\cos \phi \hat{a}_\rho - \sin \phi \hat{a}_\phi$	$\cos \phi (\sin \theta \hat{a}_r + \cos \theta \hat{a}_\theta) - \sin \phi \hat{a}_\phi$	
	$\hat{a}_y$	$\sin \phi \hat{a}_\rho + \cos \phi \hat{a}_\phi$	$\sin \phi (\sin \theta \hat{a}_r + \cos \theta \hat{a}_\theta) + \cos \phi \hat{a}_\phi$	
	$\hat{a}_z$	$\hat{a}_z$	$\cos \theta \hat{a}_r - \sin \theta \hat{a}_\theta$	

Cylindrical		ρ	ϕ	z
 $\vec{r} = (r, \phi, z)$		$\sqrt{x^2 + y^2}$	$\text{atan}\left(\frac{y}{x}\right)$	z
	$\oplus$	$r \sin \theta$	ϕ	$r \cos \theta$
	$d\vec{\ell}$	$d\rho \hat{a}_\rho + \rho d\phi \hat{a}_\phi + dz \hat{a}_z$		
	dV	$\rho d\rho d\phi dz$		
		Cartesian	Spherical	
	$\hat{a}_\rho$	$\cos \phi \hat{a}_x + \sin \phi \hat{a}_y$	$\sin \theta \hat{a}_r + \cos \theta \hat{a}_\theta$	
	$\hat{a}_\phi$	$-\sin \phi \hat{a}_x + \cos \phi \hat{a}_y$	$\hat{a}_\phi$	
	$\hat{a}_z$	$\hat{a}_z$	$\cos \theta \hat{a}_r - \sin \theta \hat{a}_\theta$	

Spherical	$\oplus$	r	θ	ϕ
 $\vec{r} = (R, \theta, \phi)$		$\sqrt{x^2 + y^2 + z^2}$	$\text{atan}\left(\frac{\sqrt{x^2 + y^2}}{z}\right)$	$\text{atan}\left(\frac{y}{x}\right)$
		$\sqrt{\rho^2 + z^2}$	$\text{atan}\left(\frac{\rho}{z}\right)$	ϕ
	$d\vec{\ell}$	$dr \hat{a}_r + r d\theta \hat{a}_\theta + r \sin \theta d\phi \hat{a}_\phi$		
	dV	$r^2 \sin \theta dr d\theta d\phi$		
		Cartesian	Cylindrical	
	$\hat{a}_r$	$\sin \theta (\cos \phi \hat{a}_x + \sin \phi \hat{a}_y) + \cos \theta \hat{a}_z$	$\sin \theta \hat{a}_\rho + \cos \theta \hat{a}_z$	
	$\hat{a}_\theta$	$\cos \theta (\cos \phi \hat{a}_x + \sin \phi \hat{a}_y) - \sin \theta \hat{a}_z$	$\cos \theta \hat{a}_\rho - \sin \theta \hat{a}_z$	
	$\hat{a}_\phi$	$-\sin \phi \hat{a}_x + \cos \phi \hat{a}_y$	$\hat{a}_\phi$	

Integrals

Line / Contour Integrals

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Line integral is an integral used to accumulate a quantity along a curve

$$\int_C f \, ds$$

→ For each arc length piece ds , take the value of f there, then add all these little amount $f \, ds$

line parameterisation

$$ds = \left| \frac{d\vec{r}}{dt} \right| dt$$

Arc length is the simplest line integral by setting $f = 1$, we are essentially just summing up all the little pieces of arc length ds $[S = \int_C 1 \, ds]$

For ECE 3121 notation, replace ds with $d\ell$.

$$\int_C f \, d\ell$$

For questions, replace $d\ell$ with the respective differential depending on the coordinate system. [Refer to Quick Glance Tables, pg above]

Example

Find line integral around a circle of radius a

$$\text{Circle: } x^2 + y^2 = a^2$$

$$\text{Line Integral} = \oint_C f \, d\ell$$

$$\text{Let } f = 1 \text{ to get pure length} \\ \Rightarrow L = \int_0^{2\pi} 1 \, d\ell \text{ --- ①}$$

$$\text{②} \rightarrow \text{①}$$

$$L = \int_0^{2\pi} \rho \, d\phi \\ = \rho [\phi]_0^{2\pi} \\ = 2\pi \rho \quad \#$$

By using cylindrical coordinate system,

$$d\ell = \overset{0}{d\rho} + \rho d\phi + \overset{0}{dz} \quad \left[\begin{array}{l} \text{To move in a circle, we only have} \\ \text{to move tangentially from the circle, that is } d\phi \end{array} \right] \\ = \rho d\phi \text{ --- ②}$$

Quick Guide to determine which coordinate system to use

Rectangular / Planar $\rightarrow$ Cartesian

Symmetry around y-axis $\rightarrow$ Cylindrical

Symmetry around a point $\rightarrow$ Spherical

Surface Integrals

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In scalar field : $\iint_S f \, dS$ Tiny curved squares

When $f = 1$, Surface Integral = Surface Area

In vector field (Flux) : $\iint \vec{F} \cdot \vec{n} \, dS$

Volume Integrals [Triple Integral]

ENG 2005 Recap

$$\iiint_V f \, dV$$

If $f = 1$, Volume Integral = Physical Volume

Electrostatic

Electrostatics is the study of electric charges and the electric field they produce when the charge configuration is NOT changing with time.

Terminologies

Source charge, Q : Responsible for creating the E-field

Test charge, q : ^{Does not disturb E-field by source charge} Sufficiently small positive charge placed somewhere in the field to "test" the force there

Point charge : Model in which the physical size of the charged object is negligible compared with the distance involved

E-field Intensity / Strength, $\vec{E}$

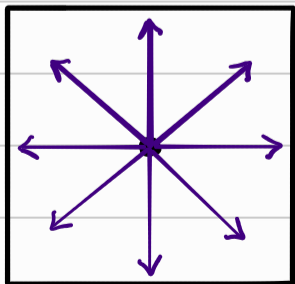
Electric Force per unit positive test charge at a point [Local Quantity]

$$\boxed{\vec{E} = \vec{F}/q} \quad (\text{OR}) \quad \boxed{\vec{F} = q\vec{E}}$$

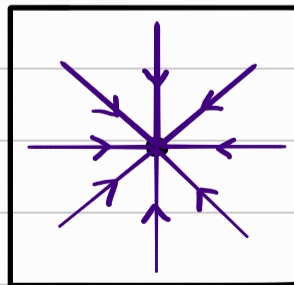
Imagine the E-field at one point is

$$\vec{E} = 5 \text{ N/C}$$

This means : A $+1\text{C}$ charge placed there would experience a 5N force in the direction of $\vec{E}$



$+Q$



$-Q$

- E-field lines start on +ve charges, ends on -ve charges
- Density of E-field lines indicate the local E-field strength

Coulomb's Law

Coulomb's Law tells us the electric force experienced by a particular test charge, q placed in an E-field

$$\vec{F} = \frac{k Q q}{R^2} \hat{a}_R \quad \text{where } k = \frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ Nm}^2/\text{C}^2$$

Magnitude Direction

By substituting Coulomb's Law into $\vec{E} = \frac{\vec{F}}{q}$

$$\vec{E} = \frac{k Q}{R^2} \hat{a}_R \quad \text{where } k = \frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ Nm}^2/\text{C}^2$$

Magnitude Direction

Take note:

$\vec{E}$ only depends on the Source charge, Q not test charge, q .

Electric Flux, ϕ_E

Electric Flux gives a measure of how much E-field passes through a surface.

$$\phi_E = \int_S \vec{E} \cdot d\vec{A}$$

Normal Vector
to Surface

Electric Flux Density, $\vec{D}$

Amount of Electric Flux, ϕ_E packed per unit area

$$D = \frac{d\phi_E}{dA}$$

Charge Distribution

Charges aren't only concentrated at a point, but charges can instead be spread along a line, a surface, a volume

Charge Density Differentials, dQ

a) Line Charge Density, ρ_l

Amount of charge per unit length

$\rho_l = 2 \text{ C/m}$: Every metre of line contains 2 C.

$\therefore$ It follows that for every tiny length, dl it contains a tiny amount of charge, dQ respective to its charge density, ρ_l

$$dQ = (\rho_l) dl$$

b) Surface Charge Density, ρ_s

Amount of charge per unit length

$\rho_s = 2 \text{ C/m}^2$: Every metre^2 of area contains 2 C

$\therefore$ It follows that for every tiny area, dS it contains a tiny amount of charge, dQ respective to its charge density, ρ_s

$$dQ = (\rho_s) dS$$

c) Volume Charge Density, ρ

Amount of charge per unit Volume

$\rho_Q = 2 \text{ C/m}^3$: Every metre^3 of Volume contains 2 C

$\therefore$ It follows that for every tiny Volume, dV it contains a tiny amount of charge, dQ respective to its charge density, ρ

$$dQ = (\rho) dV$$

Discrete Multiple Charges [Superposition]

Suppose several source charges are present, and each source charge produces its own field at point P:

$$\vec{E}_1, \vec{E}_2, \vec{E}_3, \dots, \vec{E}_N$$

The total field will be

$$\vec{E}_{\text{total}} = \sum_{k=1}^N \vec{E}_k$$

This is called the principle of superposition. However, we should only sum up matching direction (components).

Example

$$\vec{E}_1 = 3 \hat{a}_x + 2 \hat{a}_y ; \vec{E}_2 = -1 \hat{a}_x + 4 \hat{a}_y$$

$$\left. \begin{array}{l} E_{x,\text{total}} = 3 + (-1) = 2 \\ E_{y,\text{total}} = 2 + 4 = 6 \end{array} \right\} \vec{E}_{\text{total}} = 2 \hat{a}_x + 6 \hat{a}_y \neq$$

So more specifically ...

$$\vec{E} = E_x \hat{a}_x + E_y \hat{a}_y + E_z \hat{a}_z$$

$$E_{x,\text{total}} = \sum_k E_{x,k}$$

$$E_{y,\text{total}} = \sum_k E_{y,k}$$

$$E_{z,\text{total}} = \sum_k E_{z,k}$$

Continuous Charge Distributions

Realistically, charge exists continuously along the whole object.

For discrete, we used $\sum_k$, so it follows that the continuous version of summation is integration, $\int$

Tiny field produced by tiny point charge, dQ

$$\vec{E}_{\text{total}} = \sum_{k=1}^N \vec{E}_k \xrightarrow{\text{Continuously}} \vec{E}_{\text{total}} = \int_L d\vec{E} \quad \text{OR} \quad \vec{E}_{\text{total}} = \iint_S d\vec{E} \quad \text{OR} \quad \vec{E}_{\text{total}} = \iiint_V d\vec{E}$$

Discrete 1D 2D 3D

For an object that has a continuous charge distribution, we mentally chop the charged object into extremely tiny pieces, dQ , small enough that we can treat it like a tiny point charge.

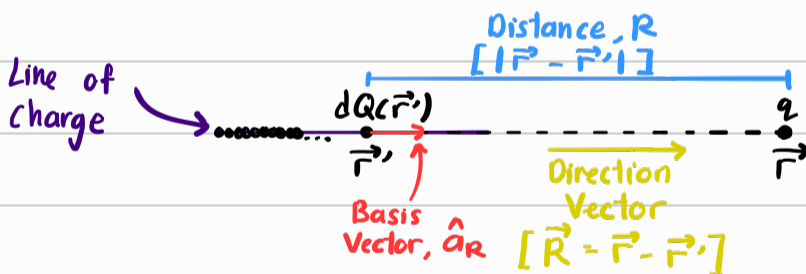
$$\vec{E} = k \left(\frac{Q}{R^2} \right) \hat{a}_R \longrightarrow d\vec{E} = k \left(\frac{dQ}{R^2} \right) \hat{a}_R$$

One point charge One tiny piece treated as point charge

Distance to test charge / Observation point Basis vector toward test charge

Let $\vec{r}$ be the fixed observation point

$\vec{r}'$ be a variable for the current position of the current dQ
[e.g. $\vec{r}' = x' \hat{a}_x + y' \hat{a}_y + z' \hat{a}_z$]



$$\text{Basis Vector, } \hat{a}_R = \frac{\vec{R}}{R}$$

$$\hat{a}_R = \frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|}$$

One particular tiny source, dQ produces

$$d\vec{E} = k \left[\frac{dQ(\vec{r}')}{|\vec{r} - \vec{r}'|^2} \right] \left[\frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|} \right]$$

$$d\vec{E} = k \left[\frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|^3} \right] dQ(\vec{r}')$$

$$d\vec{E} = k \left[\frac{\vec{R}}{R^3} \right] dQ(\vec{r}')$$

[Continue Next Pg]

Now, depending on where the charge is allowed to exist, we use different integrals as stated on the start of the previous pg.

a) Charge along a line $[dQ(\vec{r}') = \lambda(\vec{r}') dl']$ charge density differentials [pg 14 - 15]

$$\vec{E}_{\text{total}} = k \int_L \frac{(\vec{r} - \vec{r}') \lambda(\vec{r}')}{|\vec{r} - \vec{r}'|^3} dl'$$

Replace with dx' OR dy' OR dz'

b) Charge across a surface $[dQ(\vec{r}') = \sigma(\vec{r}') dS']$ charge density differentials [pg 14 - 15]

$$\vec{E}_{\text{total}} = k \iint_S \frac{(\vec{r} - \vec{r}') \sigma(\vec{r}')}{|\vec{r} - \vec{r}'|^3} dS'$$

Replace with
xy-plane: $dx' dy'$
xz-plane: $dx' dz'$
...

c) Charge throughout a volume $[dQ(\vec{r}') = \rho(\vec{r}') dV']$ charge density differentials [pg 14 - 15]

$$\vec{E}_{\text{total}} = k \iiint_V \frac{(\vec{r} - \vec{r}') \rho(\vec{r}')}{|\vec{r} - \vec{r}'|^3} dV'$$

Replace with $dx' dy' dz'$

Example Question 1

An infinitely long straight line of positive charge lies along the z -axis and has a constant line charge density ρ_l [C/m].

Find the electric field, $\vec{E}$ at an observation point P located a perpendicular distance ρ from the z -axis.

a) Express the charge element dQ and geometry

Charge Density = ρ_l

$$\therefore dQ(\vec{r}') = \rho_l dz' \quad \# \quad \text{--- (1)}$$

The observation point P is a radial distance, ρ away from z -axis

$$\Rightarrow \vec{r} = \rho \hat{a}_\rho \quad \#$$

The current charge element, dQ lies on the z -axis

$$\Rightarrow \vec{r}' = z' \hat{a}_z \quad \#$$

$$\therefore \text{Direction vector, } \vec{R} = \vec{r} - \vec{r}' = \rho \hat{a}_\rho - z' \hat{a}_z \quad \# \quad \text{--- (2)}$$

$$\Rightarrow \text{Magnitude, } R = \sqrt{(\rho)^2 + (z')^2} \quad \# \quad \text{--- (3)}$$

2) Find the field contribution $d\vec{E}$ from one tiny charge element, dQ

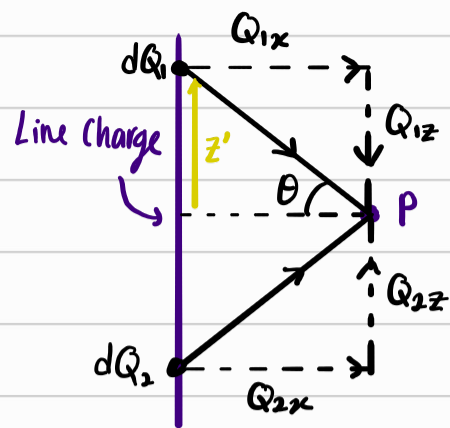
$$\text{Using } d\vec{E} = k \left[\frac{\vec{R}}{R^3} \right] dQ(\vec{r}')$$

$$\Rightarrow d\vec{E} = \frac{1}{4\pi\epsilon_0} \left[\frac{(\rho \hat{a}_\rho - z' \hat{a}_z)}{(\rho^2 + z'^2)^{3/2}} \right] (\rho_l dz') \quad \text{using (2) \& (3)}$$

$$d\vec{E} = \frac{\rho_l}{4\pi\epsilon_0} \left[\frac{(\rho \hat{a}_\rho - z' \hat{a}_z)}{(\rho^2 + z'^2)^{3/2}} \right] dz' \quad \# \quad \text{--- (4)}$$

[Continue Next Pg]

3) Use symmetry to determine which components cancel



Notice that the z-components cancel, but radial components add

$\therefore$ Final field will only be

$$\vec{E} = E_\rho \hat{a}_\rho$$

4) Integrate along the entire infinite line to obtain $\vec{E}$

From (4),

$$\begin{aligned} d\vec{E} &= \frac{\rho_L}{4\pi\epsilon_0} \left[\frac{(\rho \hat{a}_\rho - \cancel{z' \hat{a}_z})}{(\rho^2 + z'^2)^{3/2}} \right] dz' \\ &= \frac{\rho_L}{4\pi\epsilon_0} \left[\frac{\rho \hat{a}_\rho}{(\rho^2 + z'^2)^{3/2}} \right] dz' \end{aligned}$$

$$\begin{aligned} \text{Line Charge : } \vec{E}_{\text{total}} &= \frac{\rho_L}{4\pi\epsilon_0} \int_{-\infty}^{\infty} \left[\frac{\rho}{(\rho^2 + z'^2)^{3/2}} \right] dz' \hat{a}_\rho \\ &= \frac{\rho_L}{4\pi\epsilon_0} \cdot 2 \int_0^{\infty} \left[\frac{\rho}{(\rho^2 + z'^2)^{3/2}} \right] dz' \hat{a}_\rho \quad [\text{symmetry}] \end{aligned}$$

$$\begin{aligned} \text{Since } \tan \theta &= \frac{z'}{\rho} \Rightarrow z' = \rho \tan \theta \\ \therefore dz' &= \rho \sec^2 \theta d\theta \end{aligned}$$

$$\begin{aligned} \text{When } z' &= 0 \Rightarrow \theta = 0 \\ z' \rightarrow \infty &\Rightarrow \tan \theta \rightarrow \infty \\ &\Rightarrow \theta \rightarrow \frac{\pi}{2} \end{aligned}$$

$$\begin{aligned} \therefore \vec{E}_{\text{total}} &= \frac{\rho_L}{4\pi\epsilon_0} \int_0^{\pi/2} \left[\frac{2\rho}{(\rho^2 + (\rho \tan \theta)^2)^{3/2}} \right] (\rho \sec^2 \theta) d\theta \hat{a}_\rho \\ &= \frac{\rho_L}{4\pi\epsilon_0} \int_0^{\pi/2} \left[\frac{2\rho (\rho \sec^2 \theta)}{[\rho^2 (1 + \tan^2 \theta)]^{3/2}} \right] d\theta \hat{a}_\rho \\ &= \frac{\rho_L}{4\pi\epsilon_0} \int_0^{\pi/2} \left[\frac{2\rho^2 \sec^2 \theta}{[\rho^2 \sec^2 \theta]^{3/2}} \right] d\theta \hat{a}_\rho \end{aligned}$$

[Continue Next Pg]

$$\vec{E}_{\text{total}} = \frac{\rho_l}{4\pi\epsilon_0} \int_0^{\pi/2} \left[\frac{2\rho^2 \sec^2 \theta}{\rho^3 \sec^3 \theta} \right] d\theta \hat{a}_\rho$$

$$= \frac{\rho_l}{4\pi\epsilon_0} \int_0^{\pi/2} \left[\frac{2}{\rho \sec \theta} \right] d\theta \hat{a}_\rho$$

$$= \frac{\rho_l}{4\pi\epsilon_0} \cdot \frac{2}{\rho} \int_0^{\pi/2} \cos \theta d\theta \hat{a}_\rho$$

$$= \frac{\rho_l}{2\pi\epsilon_0} [\sin \theta]_0^{\pi/2} \hat{a}_\rho$$

$$= \frac{\rho_l}{2\pi\epsilon_0} [1 - 0] \hat{a}_\rho$$

$$\vec{E}_{\text{total}} = \frac{\rho_l}{2\pi\epsilon_0} \hat{a}_\rho \quad \#$$

Example Question 2 [Tutorial]

Consider a ring of charge of radius a , with charge density, ρ_L centered at the origin of the xy plane.

Find the electric field at $P_h(0,0,h)$ along the z -axis.

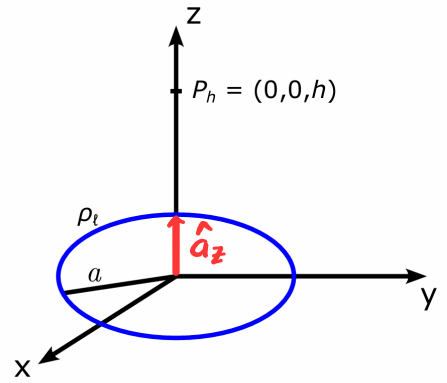


Figure: Diagram of a ring of charge at the center of the xy plane.

a) Express the charge element dQ and geometry

Charge Density = ρ_L

$$\therefore dQ(\vec{r}') = \rho_L dL$$

$$dQ = \rho_L (a d\phi) \text{ [using } s = r\theta \text{]}$$

The observation point P lies on the z -axis

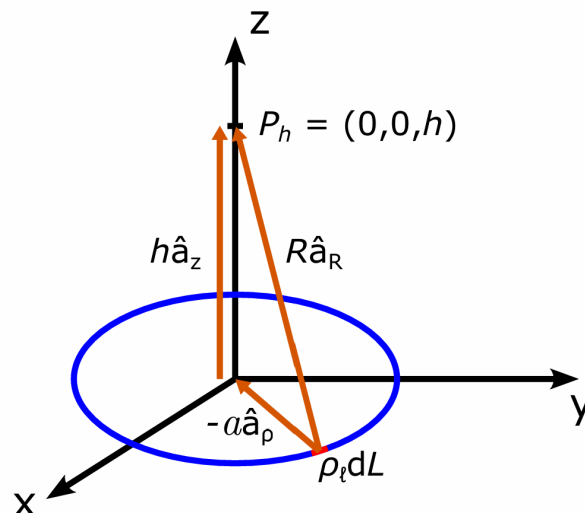
$$\Rightarrow \vec{r} = h \hat{a}_z$$

The current charge element lies on the ring

$$\Rightarrow \vec{r}' = a \hat{a}_\rho \text{ — Radial Direction towards Observation Point}$$

$$\therefore \text{Direction vector, } \vec{R} = \vec{r} - \vec{r}' = h \hat{a}_z - a \hat{a}_\rho \\ = -a \hat{a}_\rho + h \hat{a}_z \text{ — (2)}$$

$$\Rightarrow \text{Magnitude, } R = \sqrt{a^2 + h^2} \text{ — (3)}$$



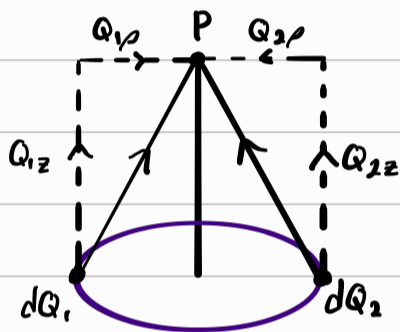
2) Find the field contribution $d\vec{E}$ from one tiny charge element, dQ

Using $d\vec{E} = k \left[\frac{\vec{R}}{R^3} \right] dQ(\vec{r})$

$$\Rightarrow d\vec{E} = \frac{1}{4\pi\epsilon_0} \left[\frac{-\alpha \hat{a}_\rho + h \hat{a}_z}{(\alpha^2 + h^2)^{3/2}} \right] (\rho \alpha d\phi) \quad \text{using (2) \& (3)}$$

$$= \frac{\rho \alpha}{4\pi\epsilon_0 (\alpha^2 + h^2)^{3/2}} (-\alpha \hat{a}_\rho + h \hat{a}_z) d\phi \quad \text{--- (4)}$$

3) Use symmetry to determine which components cancel



Note that radial components cancel,
z components add
 $\therefore$ Final field will be
 $\vec{E} = E_z \hat{a}_z$

4) Integrate along the entire infinite line to obtain $\vec{E}$

Using (4)

$$d\vec{E} = \frac{\rho \alpha}{4\pi\epsilon_0 (\alpha^2 + h^2)^{3/2}} (-\cancel{\alpha \hat{a}_\rho} + h \hat{a}_z) d\phi$$

$$= \frac{\rho \alpha}{4\pi\epsilon_0 (\alpha^2 + h^2)^{3/2}} (h \hat{a}_z) d\phi$$

$$\therefore \text{Line charge} : \vec{E}_{\text{total}} = \int_0^{2\pi} \frac{\rho \alpha}{4\pi\epsilon_0 (\alpha^2 + h^2)^{3/2}} (h \hat{a}_z) d\phi$$

$$= \frac{\rho \alpha h}{4\pi\epsilon_0 (\alpha^2 + h^2)^{3/2}} \int_0^{2\pi} d\phi \hat{a}_z$$

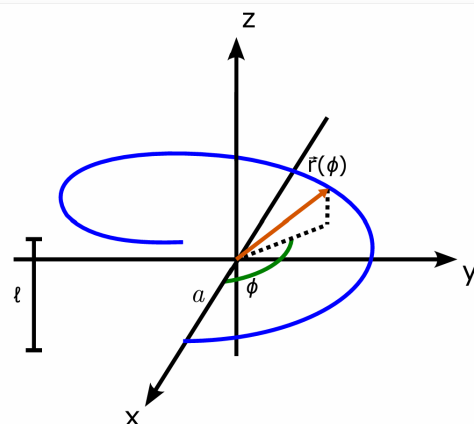
$$= \frac{\rho \alpha h}{4\pi\epsilon_0 (\alpha^2 + h^2)^{3/2}} [2\pi - 0] \hat{a}_z$$

$$\vec{E}_{\text{total}} = \frac{\overset{Q}{\rho \alpha h}}{2\epsilon_0 (\alpha^2 + h^2)^{3/2}} \hat{a}_z \quad \#$$

Example 3 [Arc Length of Helix]

A single-loop circular helix:

- has constant radius a
- rises total vertical height, l



Find total arc length L of the Helix.

$$\begin{aligned}\vec{r}(\phi) &= (a \cos \phi) \hat{a}_x + (a \sin \phi) \hat{a}_y + \left(\frac{l\phi}{2\pi}\right) \hat{a}_z \\ &= \langle a \cos \phi, a \sin \phi, \frac{l\phi}{2\pi} \rangle\end{aligned}$$

$$\text{Arc Length : } S = \int ds$$

$$= \int_0^{2\pi} \left| \frac{d\vec{r}}{d\phi} \right| d\phi$$

$$\frac{d\vec{r}}{d\phi} = \langle -a \sin \phi, a \cos \phi, \frac{l}{2\pi} \rangle$$

$$\left| \frac{d\vec{r}}{d\phi} \right| = \sqrt{a^2 \sin^2 \phi + a^2 \cos^2 \phi + \left(\frac{l}{2\pi}\right)^2}$$

$$= \sqrt{a^2 (\sin^2 \phi + \cos^2 \phi) + \left(\frac{l}{2\pi}\right)^2}$$

$$= \sqrt{a^2 + \left(\frac{l}{2\pi}\right)^2}$$

$$\therefore \text{Arc Length, } S = \int_0^{2\pi} \sqrt{a^2 + \left(\frac{l}{2\pi}\right)^2} d\phi$$

$$= \sqrt{a^2 + \left(\frac{l}{2\pi}\right)^2} \int_0^{2\pi} d\phi$$

$$= \sqrt{a^2 + \left(\frac{l}{2\pi}\right)^2} [2\pi - 0]$$

$$= 2\pi \sqrt{a^2 + \left(\frac{l}{2\pi}\right)^2}$$

$$= \sqrt{(2\pi)^2 \left[a^2 + \left(\frac{l}{2\pi}\right)^2 \right]}$$

$$S = \sqrt{4\pi^2 a^2 + l^2} \quad \#$$

Contour integrals (You do)

Problem 2: A circle of radius a is given in parametric form by

$$\vec{r}(\phi) = a \cos \phi \hat{a}_x + a \sin \phi \hat{a}_y.$$

Find the the circumference.

$$\begin{aligned} \text{Arc Length, } S &= \int ds = \int_0^{2\pi} \left| \frac{d\vec{r}}{d\phi} \right| d\phi & \therefore S &= \int_0^{2\pi} a d\phi \\ & & &= a \int_0^{2\pi} d\phi \\ & & &= a [2\pi - 0] \\ & & &S = (2\pi)a \end{aligned}$$
$$\begin{aligned} \frac{d\vec{r}}{d\phi} &= \langle -a \sin \phi, a \cos \phi \rangle \\ \left| \frac{d\vec{r}}{d\phi} \right| &= \sqrt{a^2 \sin^2 \phi + a^2 \cos^2 \phi} \\ &= \sqrt{a^2 (\sin^2 \phi + \cos^2 \phi)} \\ &= a \end{aligned}$$

Contour integrals (You do)

Problem 3: A spiral is given in parametric form by

$$\vec{r}(\phi) = \phi \cos \phi \hat{a}_x + \phi \sin \phi \hat{a}_y.$$

Find the arc length of the spiral for ϕ from $\phi_0 = 0$ to $\phi_N = \Phi$.

$$\boxed{S = \int_0^{\Phi} \left| \frac{d\vec{r}}{d\phi} \right| d\phi}$$
$$\begin{aligned} \frac{d}{d\phi} (\phi \cos \phi) &= \phi (-\sin \phi) + (1) \cos \phi \\ &= \cos \phi - \phi \sin \phi \\ \frac{d}{d\phi} (\phi \sin \phi) &= \phi (\cos \phi) + (1) (\sin \phi) \\ &= \sin \phi + \phi \cos \phi \\ \therefore \frac{d\vec{r}}{d\phi} &= \langle \cos \phi - \phi \sin \phi, \sin \phi + \phi \cos \phi \rangle \end{aligned}$$
$$\begin{aligned} \left| \frac{d\vec{r}}{d\phi} \right| &= \sqrt{(\cos \phi - \phi \sin \phi)^2 + (\sin \phi + \phi \cos \phi)^2} \\ &= \sqrt{[\cos^2 \phi - 2\phi \sin \phi \cos \phi + \phi^2 \sin^2 \phi] + [\sin^2 \phi + 2\phi \sin \phi \cos \phi + \phi^2 \cos^2 \phi]} \\ &= \sqrt{1 + 0 + \phi^2} \\ &= \sqrt{1 + \phi^2} \end{aligned}$$
$$\begin{aligned} \therefore S &= \int_0^{\Phi} \sqrt{1 + \phi^2} d\phi \\ \text{Using standard result} \\ \int \sqrt{1+x^2} dx &= \frac{1}{2} [x\sqrt{1+x^2} + \ln|x + \sqrt{1+x^2}| + C] \\ \therefore S &= \frac{1}{2} \left[\phi \sqrt{1+\phi^2} + \ln(\phi + \sqrt{1+\phi^2}) \right]_0^{\Phi} \\ &= \frac{1}{2} \left\{ \left[\Phi \sqrt{1+\Phi^2} + \ln(\Phi + \sqrt{1+\Phi^2}) \right] - 0 \right\} \\ S &= \frac{1}{2} \left[\Phi \sqrt{1+\Phi^2} + \ln(\Phi + \sqrt{1+\Phi^2}) \right] \quad \# \end{aligned}$$

